RiyalPay Product Analytics

Portfolio Project Brief

Business Context

RiyalPay is a mobile digital wallet app (P2P transfers, bill pay, top-ups, cashback) operating in Saudi Arabia and the wider Gulf. It launched in June 2025. Fourteen months in, signups are healthy, but leadership has two concerns:

1. **Activation is weak.** A large share of users complete signup but never finish KYC or make a first successful transaction — the app isn't converting registrations into real usage.
2. **An in-flight A/B test needs a rigorous read, not a gut call.** The Product team ran a 6-week experiment (`exp_kyc_streamline_2026`, Feb 2 – Mar 16, 2026) testing a shortened KYC flow (auto document scan, fewer form fields) against the existing flow, to see if it lifts KYC-approval and downstream activation. Early dashboard numbers look like a clear win for the new flow — but the VP of Product wants to know if that's trustworthy before it's rolled out to 100% of users.

Your Mandate (The Analyst Brief)

You've been brought in as the product analyst. You have four jobs:

1. **Validate the data before you trust it.** The analytics stack migrated from an in-house SDK to a Segment-based pipeline partway through this period. Confirm the event data actually matches the intended tracking plan (see the Tracking Plan document), reconcile the two export formats into one clean event table, and document what you found and fixed.
2. **Diagnose the activation funnel.** Where do users actually drop off between signup and first successful transaction? Does it differ by channel, platform, or KYC outcome? Build cohort/retention views, not just a point-in-time funnel.
3. **Analyze the KYC experiment properly.** Check assignment integrity first (don't just look at the topline lift). Test the result with appropriate rigor — significance, confidence intervals, power — and think about what could make the read misleading (novelty effects, contamination, imbalance across segments).
4. **Communicate a recommendation.** Leadership wants a go/no-go on the new KYC flow and a prioritized view of where else activation is leaking, in a form a non-technical VP can act on.

What You've Been Handed (Raw Data, As Exported By The Data Team)

- `data/users_raw.csv` — user registration records
- `data/experiment_assignments_raw.csv` — KYC-experiment assignment log
- `data/events_pre_migration_raw.csv` — product event export *before* the analytics platform migration (legacy in-house SDK format)
- `data/events_post_migration_raw.csv` — product event export *after* the migration (Segment-style format)
- `data/transactions_raw.csv` — financial transaction ledger

No cleaning has been done. This is what actually landed in the data warehouse — treat it the way you'd treat a real handoff from a data engineering team: read the tracking plan, don't assume the two event exports agree with each other or with it, and check your joins before you trust them.

Suggested Project Phases (For Your Build-Out)

1. **Data quality & implementation validation** — reconcile the two event schemas into one canonical table against the tracking plan; document every discrepancy found (naming, missing fields, timezone handling, duplicates, orphaned records) as you would in a real data-quality audit.
2. **SQL: funnel, cohort & retention analysis** — advanced SQL (window functions, CTEs) over the full event/transaction set.
3. **Python: A/B test analysis** — assignment-integrity checks, hypothesis testing, confidence intervals, power analysis, and a plain-English readout of pitfalls found.
4. **Power BI dashboard** — funnel, cohort retention, and experiment results for a product stakeholder audience.
5. **Stakeholder deliverable** — a short written recommendation (go/no-go + funnel priorities).

This document defines the ask. Deeper deliverables (SQL scripts, notebook, dashboard, written report) are the next build phases.